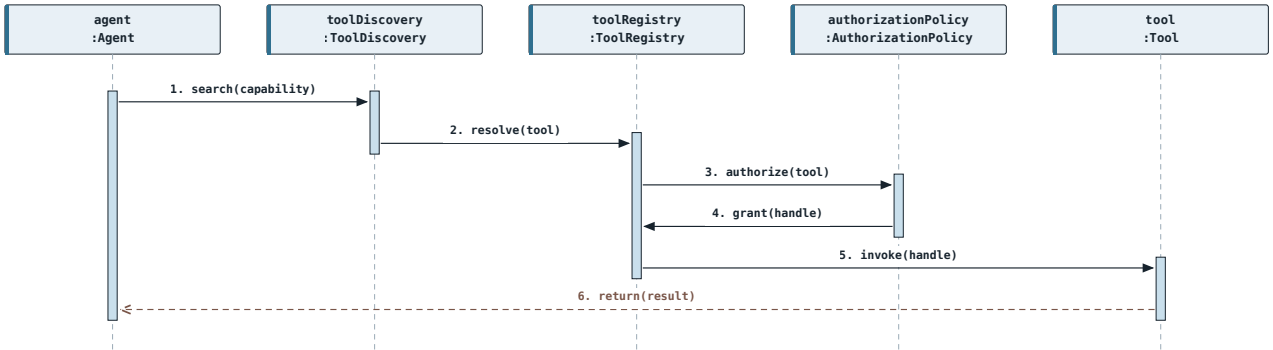
P09 Deferred tool loading

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Discover capabilities only when needed.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Available tools are too numerous or expensive to expose eagerly.
TRIGGER	The agent searches, discovers, or lazily resolves a tool.
STRUCTURE	Discovery descriptor, registry, authorization check, and invocation handle.
EFFECT	Tool schema is loaded on demand and authority is checked before execution.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Tools have discoverable metadata and a stable invocation identity.
ENSURES	A discovered tool can be invoked once with its own schema and current permissions.
INVARIANT	Lazy loading does not cache a stale authorization decision.
VIOLATION	Unknown, stale, or unauthorized tools fail explicitly before side effects.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.